Sarbajit Ghosh

CONTACT Department of CSE, IIT Kanpur

CAREER VISION My research interest aligns with the interplay between **differential privacy** and **machine learning**. I like to apply my interdisciplinary background, as well as my abilities, to find creative solutions to complex problems. This way, I like to deeply engage myself in advancing the subject to make a real-world impact. My other interests lie in the broader spectrum of **privacy-preserving technologies** and **statistics**.

ACADEMIC BACKGROUND

Ph.D (Pursuing) in Computer Science 2021 – Present
Indian Institute of Technology, Kanpur; CPI: 8.00

M.Tech in Cryptology and Information Security 2019 – 2021
Indian Statistical Institute, Kolkata; Marks: 81.35%, **First Division with Distinction**

M.Sc (Post Graduation) in Statistics 2016 – 2018
West Bengal State University, Barasat; Marks: 81.3%
Recipient of First-Class First Vice-Chancellor Gold Medal

B.Sc (Graduation) in Mathematics 2012 – 2016
Rishi Bankim Chandra College, under West Bengal State University; Marks: 54.63%

Higher Secondary Examination 2009 – 2011
Naihati Narendra Vidyaniketan, under WBCHSE; Marks: 72.6%

Secondary Examination 2009
Ramkrishna Vivekananda Mission, under WBBSE; Marks: 79.38%

GATE Qualified: Statistics (2019), Percentile: 98.01; Computer Science (2021), Percentile: 87.26

PUBLICATIONS

1. A Generalized Unknown Malware Classification. Rani, N., Mishra, A., Kumar, R., **Ghosh, S.**, Shukla, S.K. and Bagade, P. International Conference on Security and Privacy in Communication Systems (SecureComm), 2022

ACADEMIC SERVICE **Reviewer: AsiaCCS Posters**, **ACL** (2026), **FSTTCS** (2025)

TEACHING ACTIVITIES

1. **Teaching Assistant, CSE, IIT Kanpur**
   - CS729 & CS798 Differential Privacy in Machine Learning — Spring 2026, 2025
   - CS787 Generative Artificial Intelligence — Fall 2025
   - CS685 Data Mining — Fall 2024, 2021
   - CS689 Computational Linguistics for Indian Languages — Spring 2024, Fall 2022
   - CS798 Introduction to Probability for Computer Science — Fall 2023
   - ESC112 Fundamentals of Computing - II — Summer 2023
   - CS315 Principles of Database Systems — Spring 2023
   - CS657 Information Retrieval — Spring 2022
2. **Guest Lecturer, Dept. of Statistics, Chakdaha Day College** Aug 2018 – Jul 2019
   Kalyani University
   - **Responsibilities:** Teaching, Organising internal & external examinations
   - **Courses taught:** Descriptive statistics & probability theory, Statistical inference at UG level

INTERNSHIP

Statistical Programmer, iMerit Technologies, Kolkata, India Jun 2020 – Aug 2020

- **Project:** An Analysis of Factors that Impact Data Annotation Productivity in Complex CV Projects
- **Domain: Statistical Analysis**, **Socio-economic Survey**
- **Description:** To model the time per task, in CV projects of an employee, based on project characteristics and some external parameters such as training given by the company, socio-economic profile of that employee, etc as a part of a team
- **Technologies Used: R** packages **dplyr**, **ggplot2**, **caret**, **Google Sheets**, **Tableau**, NoSQL database **MongoDB**

ACADEMIC PROJECTS

1. **Internet voting using Blockchain** Spring 2021
   Guide: Dr. Souradyuti Paul, IIT Bhilai
   - **Domain: Secure Multiparty computation**, **Zero Knowledge Proofs**, **Blockchain**, **Internet voting**
   - **Description:** Application of Ethereum blockchain, in designing a complete decentralized voting protocol together with related security proofs
   - **Technologies Used: Ganache**, **Truffle**, **MetaMask**, **Solidity**, **W3.JS**
2. **A short survey on Differential Privacy** Fall 2020
   Supervisor: Dr. Srimanta Bhattacharya, R. C. Bose Centre for Cryptology and Security

3. **Secure Banking Website** Spring 2020
   Supervisor: Dr. Nilanjan Datta, ISI Kolkata
   Purpose: To create a banking website which is secure against **SQL injection**, **CSRF** and **Clickjack attack**
4. **A note on How to Access Program Counter and Instruction Register by C programming**
   Supervisor: Prof. Subhamoy Maitra, ISI Kolkata Fall 2019
5. **Robust model-based adjustments to GREG estimation when some responses are missing**
   Supervisor: Dr. Sumanta Adhya, WBSU Spring 2018
   Purpose: To study the robustness property of the GREG estimator in different cases when imputation was done by parametric and non-parametric regression and when data size is small

PROGRAMMING SKILLS
Languages: C, Python, R, HTML **OS:** Linux **Database & Publishing:** MySQL, LaTeX, Markdown
Libraries: NumPy, Pandas, Scikit-learn, PyTorch, Opacus, NLTK, spaCy

PERSONAL DETAILS
Languages: Bengali (Mother tongue), English, Hindi **Gender:** Male **Nationality:** Indian
Hobbies: Violin, Indian Classical Music, Regional History